1 Numerical Experiments

To evaluate the robustness of the proposed scheduling framework, we conducted a series of numerical experiments combining **optimization-based planning** with **stochastic simulation validation**. The experiments investigate how system performance varies with the *number of jobs* and the *robustness parameter* k.

1.1 Processing Time Model

Each job consists of two sequential stages:

- **Stage A**: manual harness alignment
- **Stage B**: automated assembly

The processing times are modeled as independent random variables:

$$T_{A,j} \sim N(\mu_{A,j}, \sigma_{A,j}^2)$$

$$T_{B,j} \sim N(\mu_{B,j}, \sigma_{B,j}^2)$$

The total processing time of job j is

$$T_j = T_{A,j} + T_{B,j}.$$

1.2 Realistic Processing Time Magnitudes

The parameter values are calibrated from an ESS assembly production dataset. Typical processing times are summarized below.

Parameter	Typical Range	Description
μ_A	76–104 min	Stage A mean time
μ_B	50–74 min	Stage B mean time
$\mu_A + \mu_B$	130–180 min	Total mean job time
σ_A	11–17 min	Stage A uncertainty
σ_B	15–22 min	Stage B uncertainty

The combined standard deviation is

$$\sigma_j = \sqrt{\sigma_{A,j}^2 + \sigma_{B,j}^2},$$

which typically lies between 20 and 25 minutes.

1.3 Robust Processing Time

To hedge against uncertainty, the planning model uses a robust processing time defined as

$$p_j^{robust} = \mu_j + k\sigma_j,$$

where k is a robustness parameter.
For an average job with

$$\mu_j \approx 150 \text{ minutes}, \quad \sigma_j \approx 22 \text{ minutes},$$

the resulting robust processing times are approximately:

k	Robust processing time (minutes)
0.5	≈ 161
1.0	≈ 172
1.5	≈ 183
2.0	≈ 194

1.4 Planning Model

The planning stage is solved using a Gurobi bin-packing formulation. Each day has a standard shift capacity of

$$C_{std} = 480 \text{ minutes}.$$

The planning horizon consists of

$$20 \text{ weekdays (four weeks)}.$$

The optimization model assigns jobs to days while minimizing overtime-related costs.

1.5 Simulation Validation

The planned schedules are validated through Monte Carlo simulation incorporating machine breakdowns. The breakdown process follows:

- Mean uptime between stops: 68.57 minutes
- Mean stop duration: 8 minutes

Each configuration is evaluated using 100 simulation replications.

1.6 Scheduling Policies

Two operational policies were evaluated.

1.6.1 Overtime Completion (Baseline Policy)

In the baseline policy, if a job cannot be completed within the regular shift, the remaining work is finished using overtime on the same day. Thus the effective daily capacity becomes

$$480 + \text{overtime}.$$

1.6.2 Weekend Extension Policy

The revised policy enforces a stricter operational rule: if the remaining time in a day is insufficient to complete a job, the job is postponed entirely to the next day. Same-day overtime completion is not allowed.

Instead, the system may utilize additional recovery capacity during weekends. The simulation horizon therefore includes:

- 20 weekday bins
- 8 weekend extension bins

Weekend production incurs both fixed and variable operating costs.

1.7 Grid Search Design

The experiments explore the following parameter combinations:

- Number of jobs: $n \in \{20, 50, 80, 100\}$
- Robustness parameter: $k \in \{0.5, 1.0, 1.5, 2.0\}$

This results in $4 \times 4 = 16$ experimental configurations.

1.8 Heatmap Results

Figure 1 summarizes the simulation outcomes across the parameter grid. The heatmaps report several performance indicators, including:

- probability of completing all jobs within the extended horizon,
- expected weekend extension cost,
- average number of weekend days used,
- average final completion day.

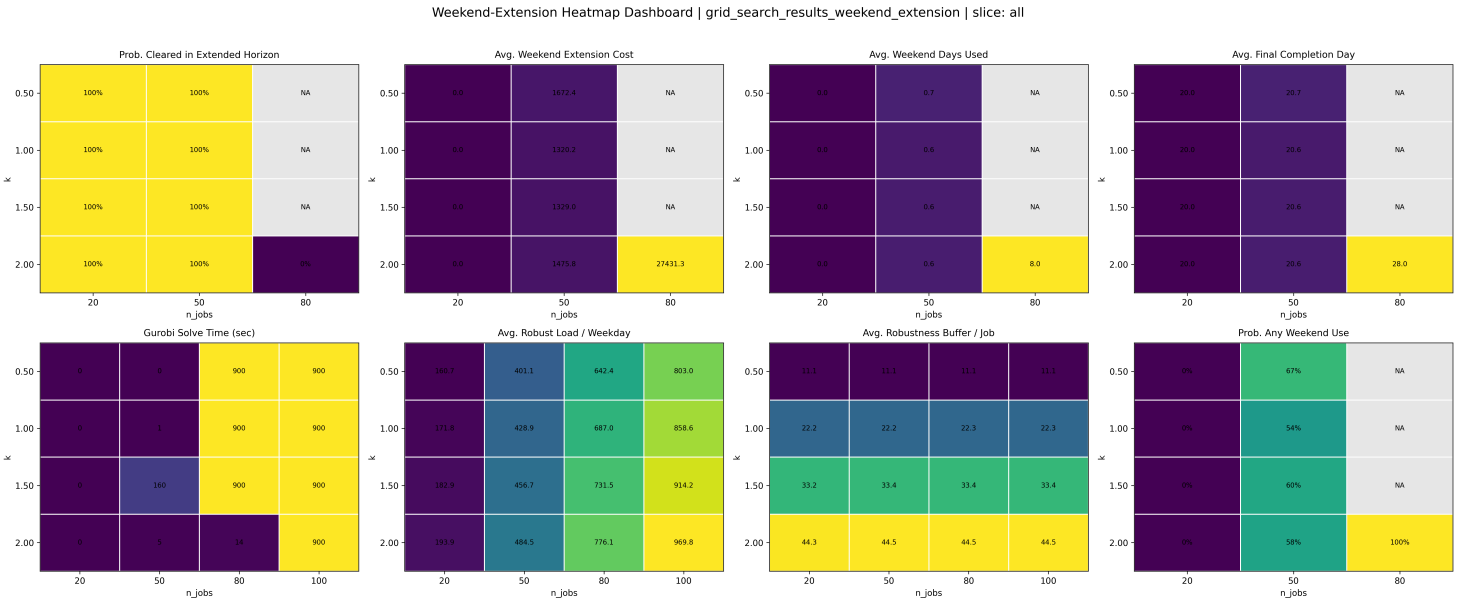

Figure 1: Heatmap dashboard summarizing the grid search results under the weekend-extension policy.

1.9 Phase Diagram

To better interpret the system behavior, the results are also summarized using a phase diagram shown in Figure 2. Each region corresponds to a different operational regime:

- Feasible within weekdays
- Feasible only with weekend extension
- Infeasible even with extension
- Computationally infeasible

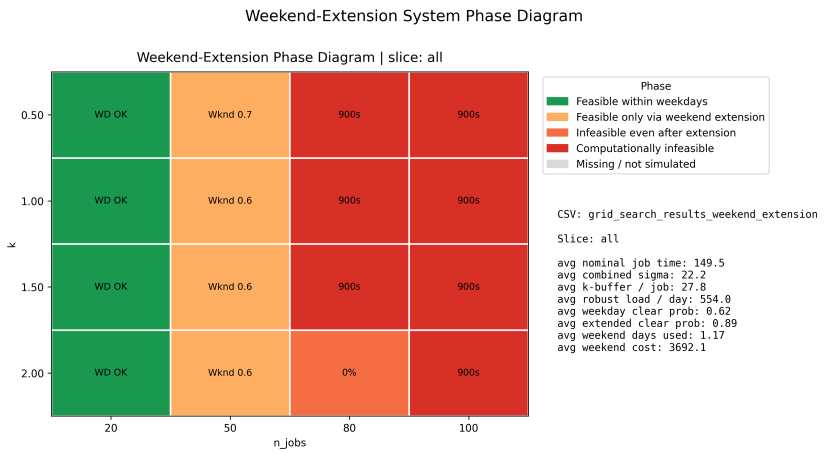

Figure 2: Phase diagram of system performance across job volume and robustness level.

1.10 Key Observations

Three distinct operational regimes emerge from the experiments.

Low-load region ($n \leq 20$). All configurations complete within the weekday horizon with no weekend usage.

Moderate-load region ($n \approx 50$). Weekday completion becomes unreliable, but weekend extension provides sufficient recovery capacity. The system remains stable with approximately 0.6–0.7 weekend days used on average.

High-load region ($n \geq 80$). The system becomes structurally overloaded. Optimization becomes computationally challenging, and simulation shows that even extended recovery capacity cannot guarantee completion.

2 Single-Run Execution Trace Visualization

To better understand how the optimized weekday schedule behaves under stochastic execution, we further examined a representative **single simulation run** using three complementary Gantt chart visualizations. Unlike the heatmaps and phase diagrams, which summarize average performance across many Monte Carlo replications, these plots provide a **sample-path view** of how jobs are actually executed over time, how backlog propagates, and how weekend extension bins are activated.

2.1 Detailed Global Gantt Chart

Figure 3 presents the detailed Gantt chart under a **global timeline**. In this figure, each execution day is split into two separate lanes corresponding to Station A and Station B. Each job therefore appears twice: once for Stage A and once for Stage B. This view is particularly useful for examining the detailed temporal structure of execution, including the propagation of delays across days and the effect of backlog jobs on downstream operations.

The main advantage of this representation is that it preserves the **continuous execution timeline** across the entire horizon. As a result, it reveals how congestion accumulates and how unfinished work pushes later tasks further into the schedule. However, because the horizontal axis is global time and the chart contains many lanes, this figure is more suitable for detailed technical analysis than for high-level presentation.

2.2 Merged Daily Gantt Chart (Shift View)

Figure 4 shows the same simulation run in a simplified **merged daily view**. In this representation, Stage A and Stage B are merged into one single block per job, so each day is displayed as one line and each job appears only once. The horizontal axis is the **within-day shift time**, which makes it easier to evaluate daily workload utilization.

This view is easier to read than the detailed global chart because it directly shows whether the realized daily workload approaches the 480-minute shift limit. It also highlights whether jobs are executed as planned or carried from backlog. In this sense, the merged shift view provides a clearer production-oriented interpretation of the schedule, especially for identifying days that are close to capacity and therefore vulnerable to spillover.

2.3 Merged Calendar Gantt Chart with Interleaved Weekends

Figure 5 presents the most intuitive visualization for managerial interpretation. The chart again merges Stage A and Stage B into one block per job, but now the days are arranged in a **calendar-style weekly structure**, where five weekdays

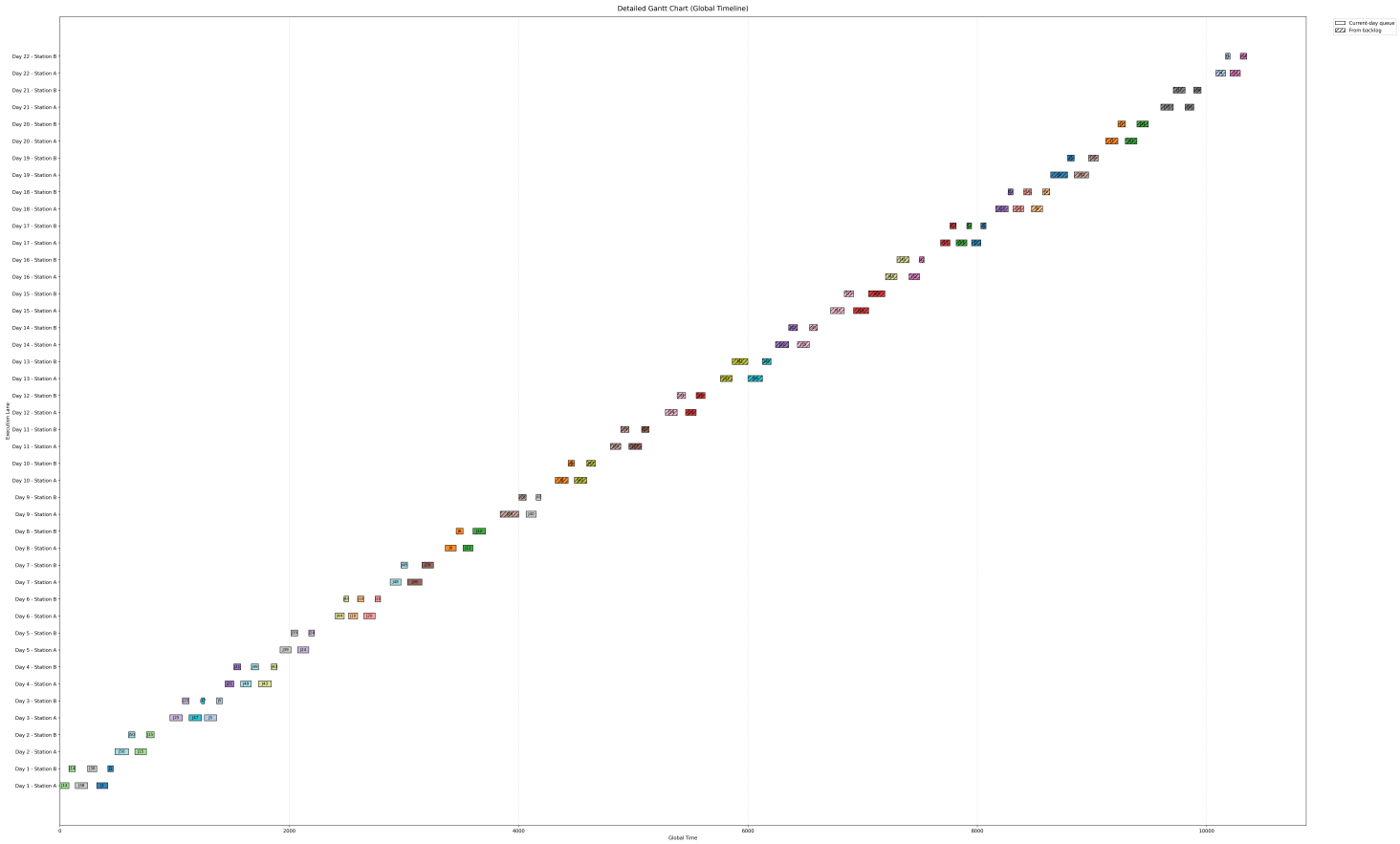

Figure 3: Detailed Gantt chart under the global timeline representation. Each job appears separately on Station A and Station B, allowing detailed inspection of execution order, backlog propagation, and cross-day delay accumulation.

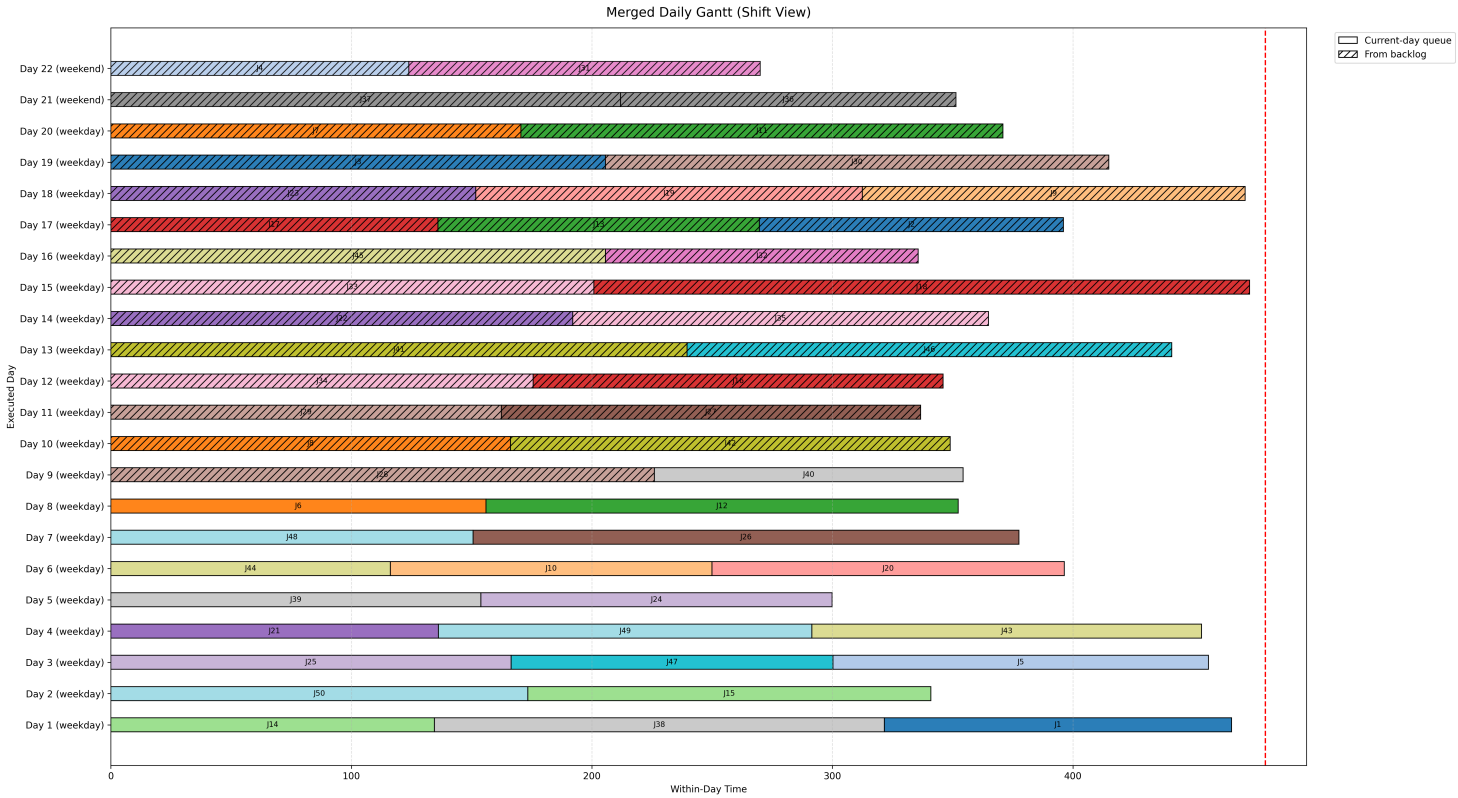

Figure 4: Merged daily Gantt chart in shift view. Stage A and Stage B are combined into one block per job, and the horizontal axis shows within-day execution time. This representation is useful for evaluating daily capacity utilization and identifying backlog-carrying jobs.

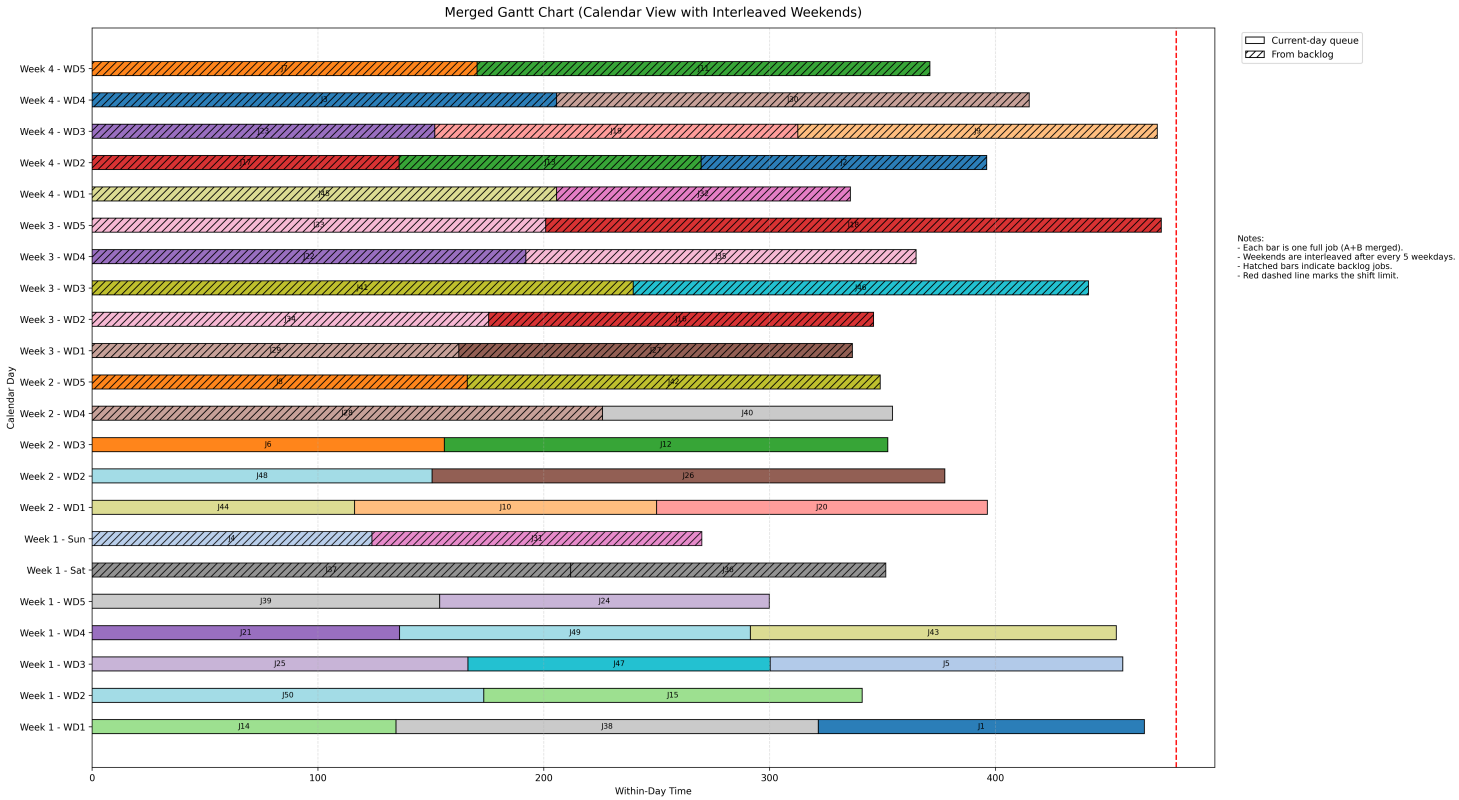

Figure 5: Merged calendar-view Gantt chart with interleaved weekends. This representation shows how weekday spillover activates weekend extension bins and how backlog propagates into the reserve recovery capacity.

are followed by two weekend positions. This allows the reader to directly observe how weekend extension bins are inserted into the production calendar.

This representation is especially useful for understanding the revised execution policy studied in this work. Under this policy, jobs that do not fit within the remaining weekday shift are deferred rather than completed with same-day overtime. Weekend bins therefore act as recovery capacity. The figure clearly shows whether weekend execution is activated, how many jobs are pushed into weekend extension days, and how backlog from late weekdays propagates into these reserve bins.

In addition to the visual Gantt representations, the full execution trace of the simulation can also be examined through the CSV output files generated during the run. These result files record the exact job execution order, start times, completion times, and whether a job originated from the current day's queue or from backlog. By inspecting these CSV logs, it is possible to trace precisely which job caused downstream delays and triggered the propagation of backlog that shifts subsequent work to later days. This provides a reproducible way to diagnose the source of schedule disruptions beyond the visual inspection of the Gantt charts.

2.4 Interpretation of the Single-Run Visualizations

Taken together, these three visualizations provide complementary perspectives on the same simulation trajectory. These plots confirm that the revised execution policy behaves as intended: jobs are not partially completed beyond the shift boundary, but are instead deferred to the next available day, potentially activating weekend recovery capacity when weekday bins are insufficient.

2.5 Remarks on Experimental Scope

It is important to note that the numerical study presented here should be interpreted as a **preliminary grid search** rather than a final calibrated industrial evaluation. The results are strongly influenced by the current assumptions on the mean and standard deviation of job processing times. In particular, the observed feasibility regions, weekend usage levels, and completion probabilities depend heavily on the chosen values of μ_A, μ_B, σ_A, and σ_B.

Although the current parameters are based on realistic industrial magnitudes, the exact outcomes may change substantially once more accurate shop-floor data become available. Therefore, the present results should be viewed as an initial stress-testing exercise designed to understand the qualitative behavior of the system, rather than as a final operational recommendation.

In future work, the job-time distributions can be recalibrated using observed production data, and the same optimization–simulation framework can then be rerun with updated parameters. This means that the methodology itself is reusable, even though the numerical values reported here may shift after calibration.

2.6 Computational Remarks

The weekend-extension experiments are also computationally intensive. A single full grid search run, including optimization and Monte Carlo simulation across all tested parameter combinations, may require several hours of computation on a standard workstation. In our experiments, one complete run typically took on the order of a few hours, and larger job sets or more replications would increase the runtime further. Consequently, while the current study is sufficient for exploratory analysis, larger-scale calibration studies will require either additional runtime or more selective experiment design.